C++14 Fundamentals - Student Edition

Week 2 - Class 3: Short, Long, and Long Long

Why This Matters

C++ offers several integer types of different widths. We already know int. Now we add short (smaller than int), long, and long long (larger).

Short

short is at least 16 bits, usually 2 bytes. Range signed: -32768..32767. Unsigned: 0..65535.
In expressions, short values are promoted to int automatically.

```
#include <iostream>
#include <limits>
using namespace std;
int main(){
    cout<<"sizeof(short)="<<sizeof(short)<<" bytes\n";
    cout<<"short min="<<numeric_limits<short>::min()<<" max="<<numeric_limits<short>::max()<<"\n";
}
```

Long and Long Long

long >=32 bits, long long >=64 bits. On Linux 64-bit: long=8 bytes. On Windows: long=4 bytes. long long always >=64 bits.

```
#include <iostream>
#include <limits>
using namespace std;
int main(){
    cout<<"sizeof(long)="<<sizeof(long)<<" bytes\n";
    cout<<"long min="<<numeric_limits<long>::min()<<" max="<<numeric_limits<long>::max()<<"\n";
    cout<<"sizeof(long long)="<<sizeof(long long)<<" bytes\n";
    cout<<"long long min="<<numeric_limits<long long>::min()<<" max="<<numeric_limits<long long>::max()<<"\n";
}
```

Unsigned Variants

Each type has unsigned variant: unsigned short, unsigned long, unsigned long long.

Common Mistakes

1. Assuming short is always 2 bytes (only guaranteed >=16 bits).
2. Forgetting short promotes to int in expressions.
3. Assuming long is always 64 bits (on Windows it is 32).
4. Forgetting unsigned doubles positive range but has no negatives.
5. Depending on built-in widths instead of <cstdint> types for fixed widths.

Exercises

1. Print size and ranges of short, long, long long.
2. Compare results across OS.
3. Show promotion of short in expressions.

Project Tie-In

Expand Type Explorer to:
- Print ranges of short, long, long long (signed/unsigned).
- Compare results with another system.
- Identify if your system uses LP64 or LLP64.
- Show short promoted in arithmetic.